



# Black Swan - R&D update

19/12/2023



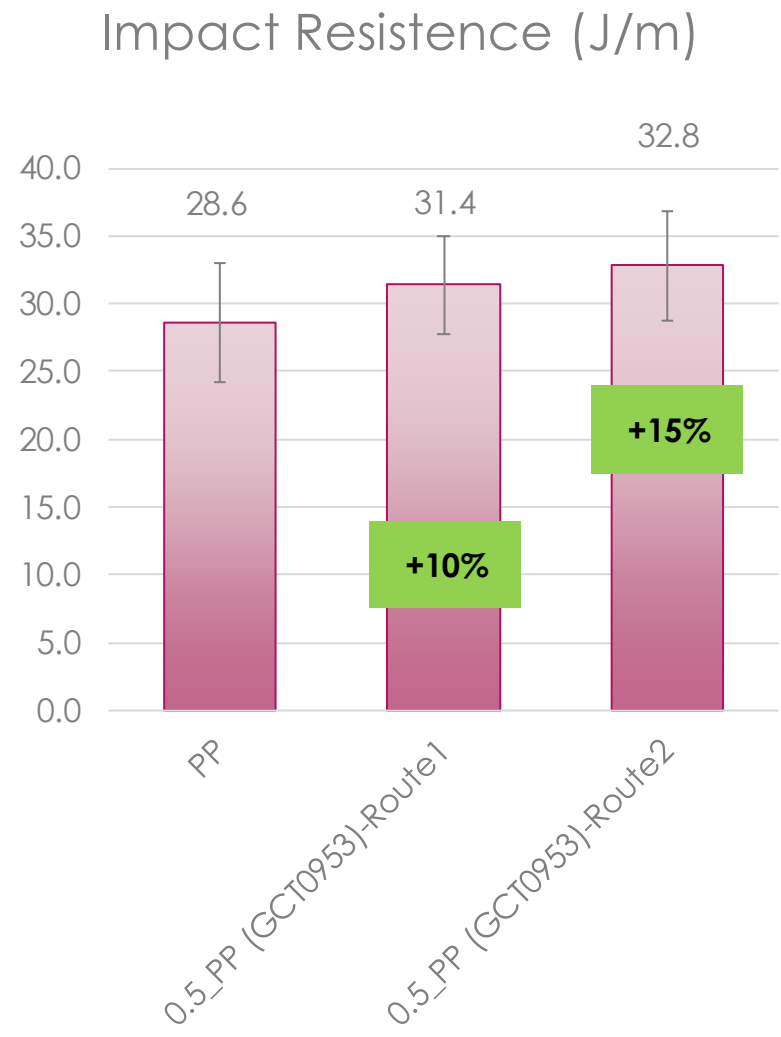
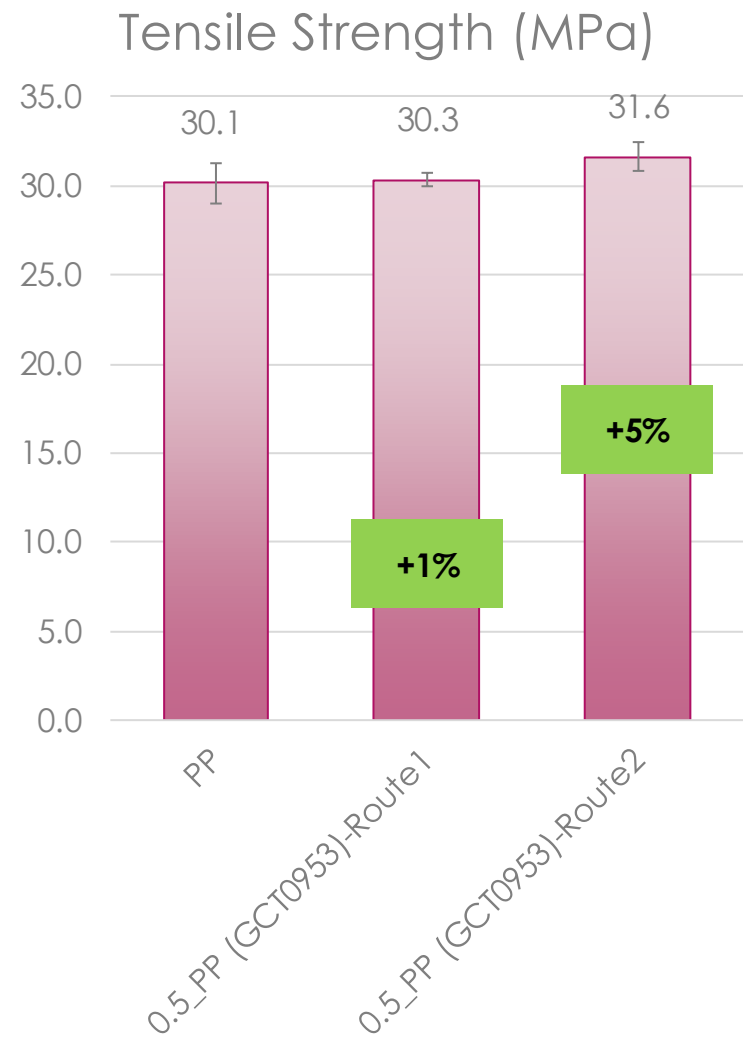
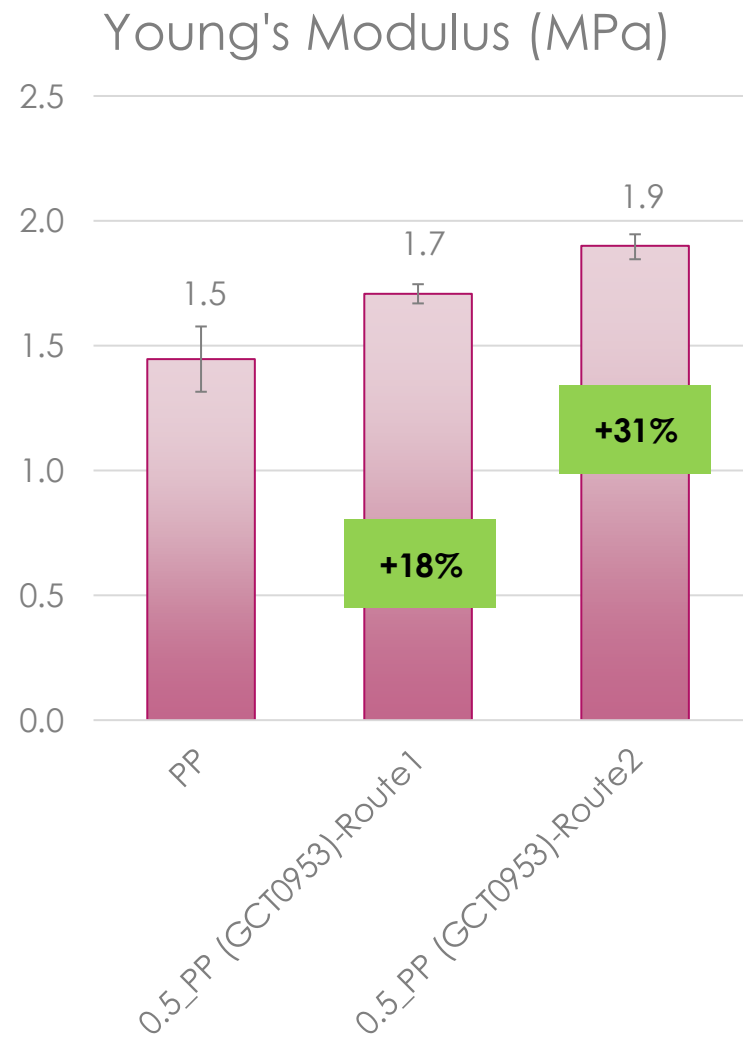
# Developments

**PP COMPOSITES  
PROCESSED BY TWO  
DIFFERENT ROUTES**

**PE COMPOSITES  
COMPARED WITH TWO  
DIFFERENT GRAPHENES**



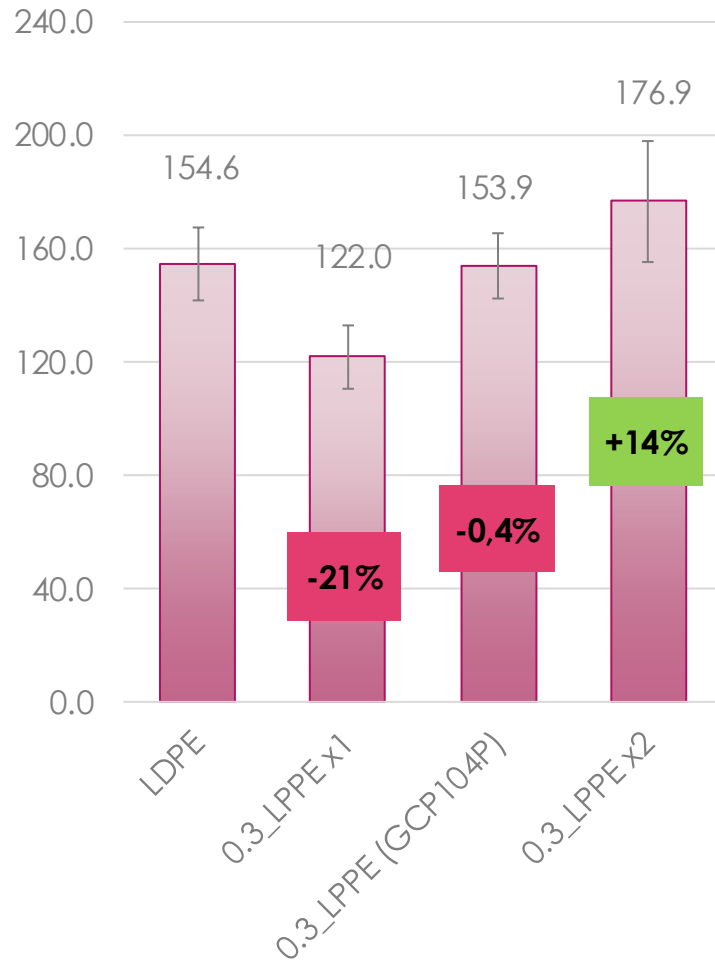
# PP Composites



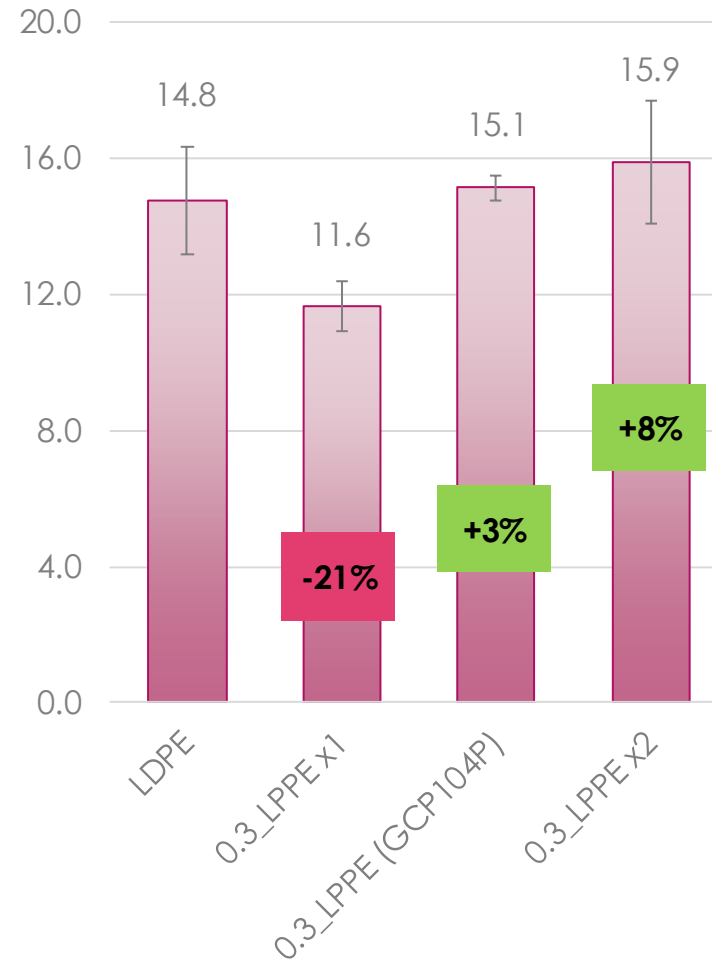
■ *Injected samples*

# PE Composites

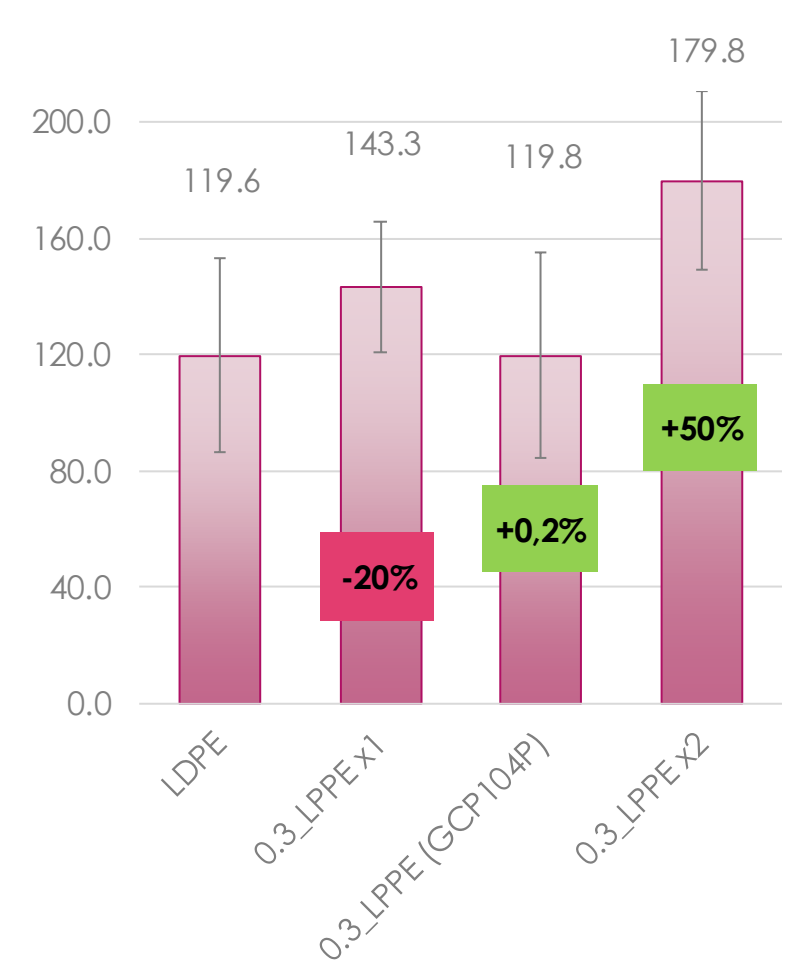
## Young's Modulus (MPa)



## Tensile Strength (MPa)



## Total Elongation (%)



- *Film samples*

# Next steps

**December:** Production of PE and PP masterbatches containing BlackSwan Graphene (BS8 and BS3);

**January-March:** Production and characterization of composites with different concentrations;

**March:** New Update.